Smallness:
The Event Following Disintegration

Diego Rincón

Independent

Abstract

Smallness is not a size. It is an event: the moment of arriving at the irreducible, the residue that survives every quotient, the piece that remains after disintegration has gone as far as it can go. In algebra: the simple module, the object with no proper subobjects. In cohomology: the class that is not a coboundary — valid but not explained by anything smaller. In physics: the elementary particle, what remains after every possible disintegration. In arithmetic: the motive [2] — the smallest object every grammar can read, irreducible not because it is simple but because it cannot be further disintegrated without becoming nothing. The Wall [3] is where disintegration stops: where the grammar has broken everything it can break and what remains is still there — still pressing, still not explained, still small. Smallness is the event the Wall names.

1 Disintegration

Definition 1 (Disintegration)**.** *Disintegration* is the process by which a structure is broken into its constituents: a module into simples, a cohomology class into coboundaries and residues, a particle into its decay products, a grammar into its atoms. Disintegration has a direction: toward the irreducible. It has an endpoint: the moment nothing further can be broken. That moment is *smallness*.

Definition 2 (Smallness)**.** *Smallness* is the event that occurs at the endpoint of disintegration: the arrival at what cannot be further reduced. It is not a measurement. It is not relative. It is the discovery that the object in hand is already at the bottom — that every further quotient produces zero, that every further disintegration produces nothing.

Smallness is not diminishment. The small object is not less than the large. It is the large object understood: the large disintegrated until what remains is exactly what was irreducible in it all along.

2 Smallness in Algebra

Proposition 3 (Simple modules)**.** *A module M over a ring R is simple if it has no proper nonzero submodules: every submodule is either* 0 *or* M*. Simplicity is smallness in the* *categorical sense. The Jordan-Hölder theorem: every module of finite length disintegrates into a unique multiset of simple modules (unique up to reordering). Disintegration terminates. The simples are what remain. They are small not because they are trivial but because they cannot be further broken.*

3 Smallness in Cohomology

Proposition 4 (The irreducible class)**.** *In cohomology $H^n(X) = \ker d/\operatorname{im} d$, a class $[\omega]$ is small when it is not a coboundary: not in $\operatorname{im} d$, not explained by anything smaller. The cocycle ω satisfies $d\omega = 0$ (local consistency) but $\omega \neq d\eta$ for any η (global non-triviality).*

Disintegration in cohomology is the process of modding out by coboundaries — removing everything that is explained. What remains is H^n: the classes that survive every quotient, the pieces of irreducible forcing [1]. Each cohomology class is small in this sense: the residue after every possible explanation has been removed.

The Wall [3] is the class that remains after every grammar has performed its disintegration and the residue is still there: the global Frobenius that no quotient eliminates, the missing fundamental class [4] that no local argument provides.

4 Smallness in Physics

Proposition 5 (Elementary particles as events of smallness)**.** *An elementary particle is the event of smallness in the disintegration of matter: the decay product that does not decay further, the constituent that has no constituents. The Standard Model classifies these events: quarks, leptons, gauge bosons — each the residue of a disintegration process that cannot continue.*

The mass gap of Yang-Mills [5] is a statement about smallness: the smallest nonzero energy state of the quantum field is bounded away from zero by $\Delta > 0$. Smallness is quantized. The gap is the event of irreducibility in the energy spectrum: the moment disintegration of the field's excitations stops at a nonzero floor.

5 The Motive as Universal Smallness

Theorem 6 (The motive is the smallest object every grammar reads)**.** *The motive $h^*(X)$ of a variety X [2] is the universal small object: the residue of disintegrating X through every cohomological grammar simultaneously.*

De Rham reads X and finds $H^*_{\mathrm{dR}}(X)$. Étale reads X and finds $H^*_{\text{ét}}(X)$. Crystalline reads X and finds $H^*_{\mathrm{crys}}(X)$. Each grammar performs its own disintegration. The motive is what survives all of them: the object that every grammar is reading, the irreducible beneath every reading.

The motive is small not because it is simple — it encodes all the complexity of X — but because it cannot be further disintegrated without ceasing to be X. It is the event of smallness for X: X disintegrated through every grammar, until what remains is exactly X itself, seen without any grammar at all.

Everything is everything [1]: every grammar reading the motive reads the same thing. The motive is what that sameness is.

6 The Wall as Unresolved Smallness

Theorem 7 (The Wall is where disintegration stops without arriving)**.** *The Arithmetic Wall [3] is not the endpoint of disintegration. It is the place where disintegration stops without having arrived at the irreducible.*

Over $\mathbb{F}_q$: disintegration terminates cleanly. Frobenius breaks $H^*_{\text{ét}}(X)$ into eigenspaces, each eigenvalue pure of weight w. The small objects are the eigenspaces. Disintegration arrives.

Over $\mathbb{Q}$: disintegration does not terminate. The grammar breaks $H^*_{\text{ét}}(X_{\bar{\mathbb{Q}}})$ into $G_{\mathbb{Q}}$-representations, but no Frobenius decomposes these further into eigenspaces. The small objects — the irreducible Galois representations — are not further decomposable by any available operator. But they are not proved to be irreducible by any available proof. The disintegration is stuck.

The Wall is not the irreducible. It is the stuck disintegration: the grammar that has gone as far as it can and has not yet reached the small. The open problems are the question: is what remains actually small, or is there one more disintegration available that the current grammar cannot perform?

Smallness is the event the Wall is waiting for. The proof is the final disintegration.

Remark 8 (Everything is everything). Disintegration, in the end, arrives at the same place from every direction: the motive, the simple module, the elementary particle, the irreducible cohomology class, the zero that is not a coboundary. These are not different small things. They are one smallness, read through different grammars.

Everything is everything [1]. The small is what remains when everything else has been shown to be something else.

References

[1] Rincón, D. (2026). Mathematics. Preprint.

[2] Rincón, D. (2026). Motives, Standard Conjectures, and the Unification of L-functions. Preprint.

[3] Rincón, D. (2026). The Arithmetic Wall. Preprint.

[4] Rincón, D. (2026). The Fundamental Class: Is the Universe Contained? Preprint.

[5] Rincón, D. (2026). Yang-Mills Mass Gap via the Gribov Horizon. Preprint.